

Subject: Computer Science Theory**Time allowed:** 2 hours**Maximum marks:** 50**Date:** 7th August 2026Name: _____

Instructions

- All questions are compulsory.
- The marks for each question are shown in brackets on the right.
- Write all your answers in the answer sheet provided, not on this paper.
- Show your working for conversions, truth tables and traces — the working carries marks.

Part A — Multiple Choice Questions*8 questions — 8 marks*

Tick (✓) the correct option for each question.

Q1. What is the full form of DHCP?

[1]

- (a) Domain Host Control Protocol
 - (b) Dynamic Host Configuration Protocol
 - (c) Data Host Communication Protocol
 - (d) Dynamic Hypertext Connection Protocol
-

Q2. Which pair of keyboard shortcuts is used to copy and then paste?

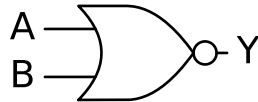
[1]

- (a) Ctrl + X, then Ctrl + V
 - (b) Ctrl + C, then Ctrl + V
 - (c) Ctrl + C, then Ctrl + P
 - (d) Ctrl + S, then Ctrl + V
-

Q3. Which of the following is **not** a valid IP address? [1]

- (a) 192.168.1.1
 - (b) 10.0.0.255
 - (c) 172.16.300.5
 - (d) 8.8.8.8
-

Q4. Which of the following is the Boolean expression for the output Y of the circuit below? [1]



- (a) $Y = \overline{A} + \overline{B}$
 - (b) $Y = \overline{A} \cdot \overline{B}$
 - (c) $Y = A + B$
 - (d) $Y = A \cdot B$
-

Q5. ASCII uses 7 bits for each character. **How many different** characters can ASCII represent? [1]

- (a) 7
 - (b) 64
 - (c) 128
 - (d) 256
-

Q6. You type **A1+A2** into a cell in LibreOffice Calc and press Enter. What appears in the cell? [1]

- (a) The sum of the values in A1 and A2
 - (b) The text **A1+A2**
 - (c) An error message
 - (d) 0
-

Q7. You need to send 50 files to your teacher as a single email attachment. Which format should you use? [1]

- (a) .png
 - (b) .jpg
 - (c) .zip
 - (d) .txt
-

Q8. What is the value of $A \cdot \overline{A}$? [1]

- (a) A
 - (b) $\overline{A}$
 - (c) 0
 - (d) 1
-

Part B — Short Answer Questions

7 questions — 8 marks

Q9. Draw the logic-gate circuit for the Boolean expression below. [1]

$$Y = A \cdot B + \overline{C}$$

Q10. The shaded cells in the spreadsheet below are selected. Write the address of this selection. [1]

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						

Q11. What is the keyboard shortcut for undoing an action in LibreOffice Writer? [1]

Q12. One pixel of an RGB image uses one **byte** for each of the colours. How many **bits** is that per pixel? [1]

Q13. What is the value of $23 // 4 - 23 \% 4$?

[1]

Q14. A computer game is an example of which type of software?

[1]

- (a) System software
- (b) Application software
- (c) Utility software
- (d) Operating system

Q15. The spreadsheet below is open in LibreOffice Calc. Write what appears in each cell after the formula is typed into it and Enter is pressed.

[2]

	A	B	C	D
1	Item	Price		
2	Pen	40		
3	Notebook	150		
4	Bag	850		
5	Water bottle	250		
6	Eraser	10		
7	Lunch box	300		
8				
9				

(i) Cell B8 holds the formula `=SUM(B2:B7)`. What does B8 show?

(ii) Cell B9 holds the formula `=SUMIF(B2:B7, ">200")`. What does B9 show?

Part C — Boolean Algebra

2 questions — 6 marks

Q16. For the Boolean expression below:

[3]

$$Y = A \cdot \overline{B} \cdot C + \overline{A} \cdot B$$

(i) Draw the complete truth table for Y .

(ii) Find the value of Y when $A = 1, B = 0, C = 1$ and when $A = 1, B = 1, C = 0$. Show the substitution.

Q17. For the Boolean expression below:

[3]

$$Y = (A + B) \cdot \overline{C}$$

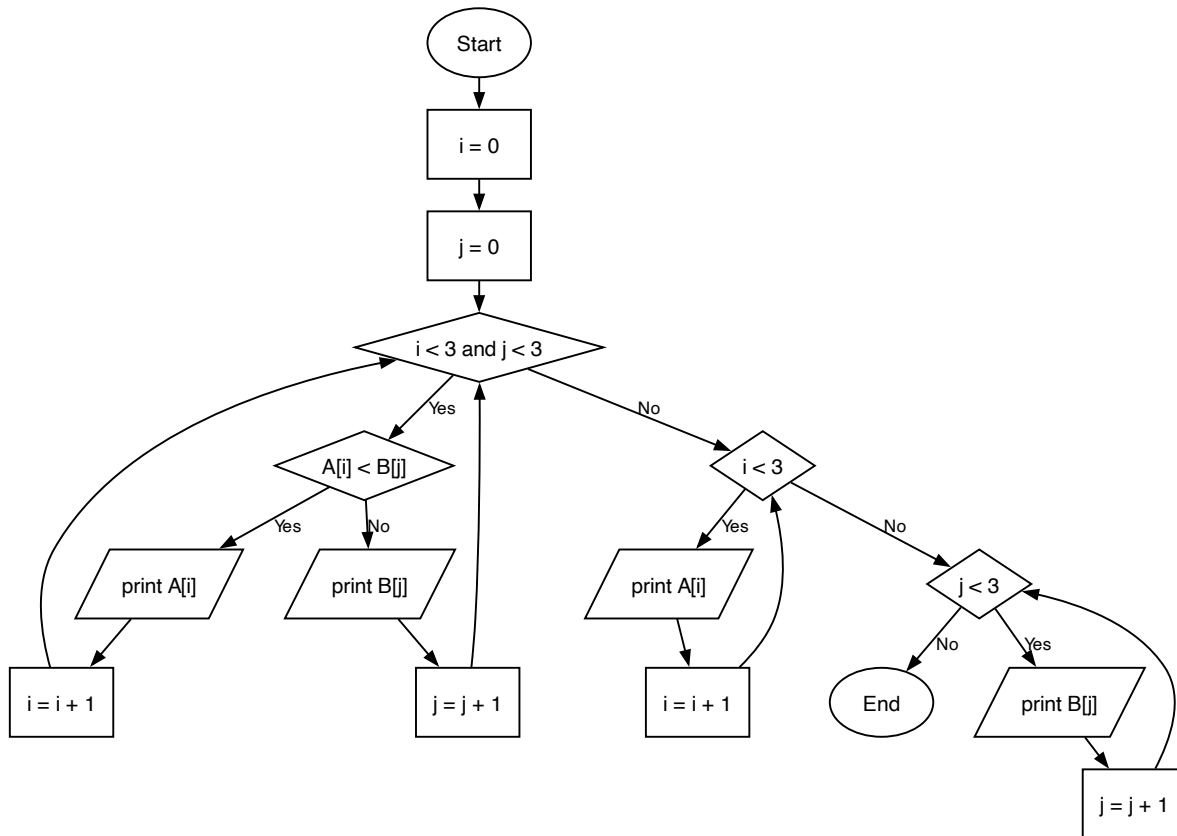
- (i) Draw the logic-gate circuit for Y .
 - (ii) Draw the truth table for the same circuit.
-

Part D — Number Systems*2 questions — 5 marks*

Q18. Convert the binary number 1011010_2 to decimal. [2]

Q19. Convert the hexadecimal number $2AF_{16}$ to binary. [3]

Q20. Trace this algorithm, taking the list A to be $[2, 5, 9]$, taking the list B to be $[3, 6, 8]$, and draw the trace table. [5]

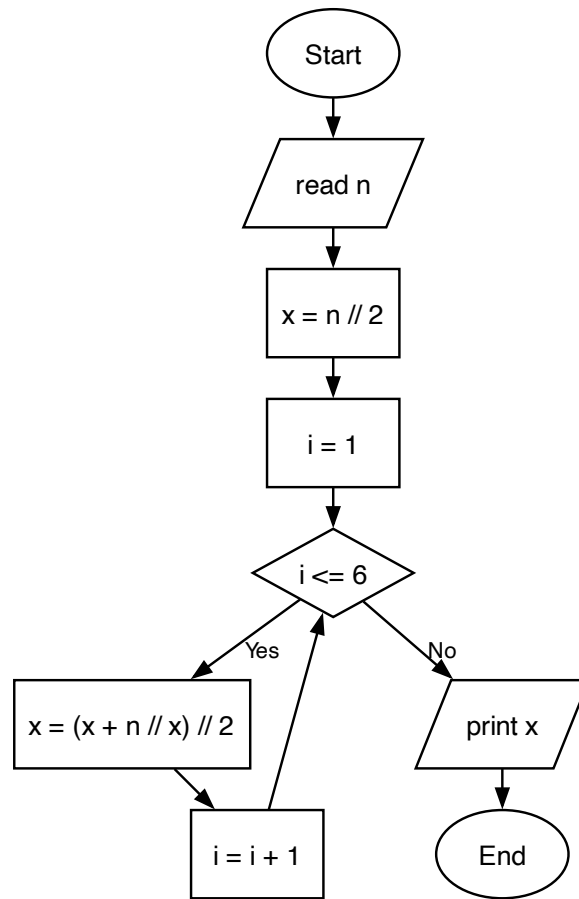


Copy this table into your answer sheet and complete it.

i	j	output

Q21. Trace this algorithm for $n = 1000$, and draw the trace table.

[4]

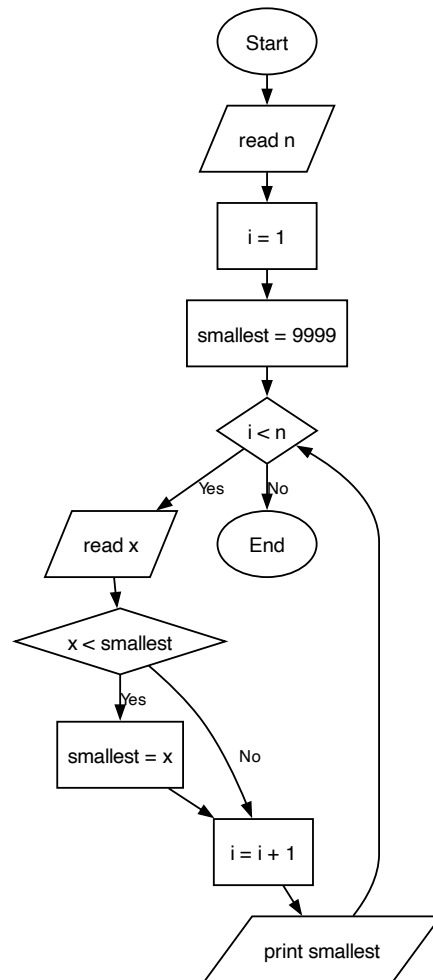


Copy this table into your answer sheet and complete it.

n	x	i	output

Q22. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

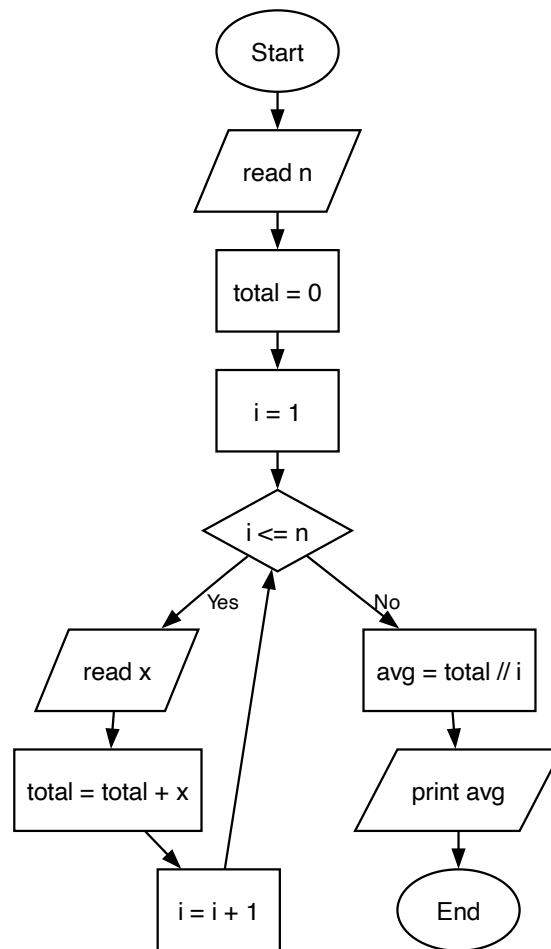
The program should read how many numbers there are, then read that many numbers, and finally print the smallest of them — once.



input	should print	actually prints
n = 5, x = 30, 25, 15, 20, 10	10	30, 25, 15, 15

Q23. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

The program should read how many numbers there are, then read that many numbers, and print their average.



input	should print	actually prints
n = 4, x = 10, 20, 30, 40	25	20
n = 5, x = 20, 30, 10, 40, 20	24	20

Q24. Read the description below, then answer the parts that follow.

[7]

A program reads a whole number n and prints the factorial of n . The factorial of n is $1 \times 2 \times 3 \times \cdots \times n$, so the factorial of 4 is $1 \times 2 \times 3 \times 4 = 24$.

- (i) Write the algorithm in words, step by step.
 - (ii) Draw the flowchart for that algorithm.
 - (iii) Trace your flowchart for $n = 5$ and $n = 6$, and write what it prints in each case.
-

Q25. Read the description below, then answer the parts that follow.

[3]

A shop bills a customer. The program reads the price of one item and the quantity bought, and works out the total as $\text{price} \times \text{quantity}$. If the total is more than 1000, the shop gives the customer a 10% discount on the total. The program then prints the amount the customer has to pay.

- (i) Write the algorithm in words, step by step.
- (ii) Draw the flowchart for that algorithm.